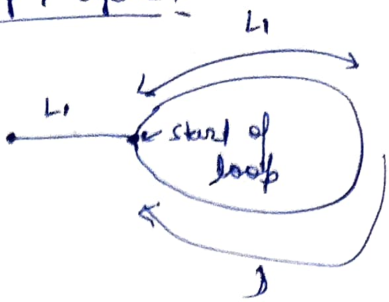
8) How Do you find the Start of loop / cycle?



- Slow moves in one step
- fast moves in 2 steps

∴ If slow moves in x distance

- Fast have moved $2x$

- Consider the distance from start to the start point of loop as L1.
- By the time slow reached start of loop L1, the fast will be $\underline{2L1}$

$$\left.\begin{array}{l} \therefore L_1 + L_2 = fast \\ L_1 = slow \end{array}\right\}$$

- Now both are in loop.

∴ what is the circle / loop length?

$\underline{M}$; $M = L_1$ (advance distance from slow).
+ d (the remaining distance from fast to slow).

- Now both are in loop;
- when slow moves & fast moves; their Distance tends to come down by '1'

ith iteration : $M = L_1 + d$ (as the Dens

$\therefore L_1 = jm - d$ D is the Distance b/w slow & fast

start =

i+1 th D ;

i: d distance

$i+1$: ~~d+1~~ $d-1$

$i+2$: $d-2$

$i+3$: $d-3$

$i+d$: $d-d=0$

∴ there is no 'd' between

~~start~~ &

slow & fast is met now.

∴ Now what is left in whole cycle ?? / Loop ??

Then: $m = L_1 + d$

Now: $d = 0$;

$\boxed{m = L_1}$

∴ what is the Distance from start ?? L_1

$\boxed{\Rightarrow L_1}$

∴ Now put $\boxed{\text{start} \Rightarrow \text{slow}}$ then start looping

∴ ~~token~~

→ Start traversing one step each

→ $i+n$: L_1

$i+1$: $L_1 - 1$

$i+2$: $L_1 - 2$

$i + L_1 + n$: $L_1 - L_1 = 0$

∴ After meeting fast & slow,

keep slow to start, now move the start & fast

one move at a time.

∴ The point when ~~start~~ & slow & fast meet is

the start of loop.